A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 14

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** (2.9 - 0.7) / 2.9 x 100 = 75.9% (accept 76%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Although the larger vehicle uses more energy in total, it carries many more people / a larger load **1** So the energy per passenger (or per tonne of goods) is lower – steel wheels on rails also have low rolling resistance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
01.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Flexibility of routes / destinations and door-to-door convenience **1** Time / speed and reliability where public transport is infrequent **1** Low fuel tax / cheap fuel **1** Carrying goods – comfort – status **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	3
01.4	**1 mark each** 1 mark for each of the 4 points below. **1** Integrated systems (park and ride, bicycles on trains, linked timetables) make it convenient to use the most efficient mode for each part of a journey **1** So more journeys shift from cars to bulk transport / cycling **1** ATM / smart motorways use variable speed limits to prevent congestion and keep traffic flowing freely **1** Vehicles avoid stop-start driving and run nearer their optimum speed (about 56 mph), reducing fuel wasted in acceleration and idling **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 225-228; AQA spec 3.3.4.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Asbestos:** fibres become airborne when it disintegrates and are inhaled, causing asbestosis (scar tissue thickens alveoli) and mesothelioma – disposal: double-wrapped in heavy-duty polythene in a specialised, recorded landfill (secure, permanent, sealed storage) **2 Cyanide:** very toxic enzyme inhibitor (aerobic respiration) if inhaled / ingested – disposal: incineration so carbon and nitrogen are separated and oxidised to CO2 and NOx (much smaller problem) **2 High-level radioactive waste:** intense ionising radiation, heat generation, long half-lives – disposal: vitrification - mixed with molten glass in sealed stainless steel containers in ventilated concrete stores **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **incineration** → burning waste, usually with energy recovery • **ionising radiation** → radiation with enough energy to remove electrons and form ions • **vitrification** → sealing waste into glass **Do not credit** • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	6
02.2	1 mark for the point below. **1** The fibres are bound / cannot become airborne so cannot be inhaled **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Waste is mixed with a cement slurry and poured into containers of impermeable, unreactive material **1** The solid cement immobilises the waste so it cannot leak / leach out **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **impermeable** → does not let water through **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
02.4	1 mark for the point below. **1** Even if the glass shatters the waste stays encapsulated in the fragments – glass is more stable / resistant to heat and leaching over very long timescales **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **vitrification** → sealing waste into glass • **leaching** → nutrients or ions washed down out of the soil / washed away by water	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 284-285, 303; AQA spec 3.4.3.2.12.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** Nuclei of large atoms (uranium-235, plutonium-239) split when hit by a neutron **1** A small amount of mass is converted to a large amount of energy ($E = mc^2$) and more neutrons are released, causing a chain reaction (controlled by control rods / moderator) **1** The heat boils water to steam which drives turbines and generators **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **control rods** → absorb neutrons to control the rate of fission • **moderator** → slows the neutrons so that fission continues **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	3
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Output changes slowly (days) so it cannot follow rapid fluctuations in demand **1** Very high construction cost but low fuel cost means it is most economic running continuously at full output **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	2
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Only 30 t of fuel a year (very high energy density) so no need for railways / ports to deliver millions of tonnes of coal **1** Still need condenser cooling water (coast / large lake) and a stable site **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	2
03.4	1 mark for the point below. **1** Uranium must be mined, purified, concentrated and processed into fuel using (fossil) energy – reactor construction and decommissioning have embodied energy **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **decommissioning** → safely dismantling a power station at the end of its life • **embodied energy** → all the energy used to make and transport a product **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	1

Do not write outside the box

Question	Marking guidance	Marks
03.5	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** No commercial reactor has been fully decommissioned so costs are unknown and often exceed estimates **1** Large projects often overrun original cost estimates **1** Long-term waste disposal costs uncertain **1** State subsidies may hide costs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 191-193; AQA spec 3.3.4.1.	
04.1	1 mark for the point below. **1** The time taken for half of the atoms of a radioactive isotope to decay (activity halves) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **half-life** → the time for the activity, or the amount, to fall by half **Do not credit** • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues	1
04.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 32 / 8 = 4 half-lives **1** 6400 / 2^4 = 400 Bq **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (Bq) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
04.3	**1 mark each** 1 mark for each of the 3 points below. **1** Short half-life isotopes (I-131) release all their radiation quickly: dangerous but only briefly so short-term precautions (potassium iodide tablets, evacuation) are sufficient **1** Very long half-life isotopes (U-238) emit only small amounts of radiation in any period so the level of danger is low **1** Decades-long half-lives combine a high rate of radiation release with persistence for most or all of a human lifetime, so short-term protection is inadequate (land contaminated for generations) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **half-life** → the time for the activity, or the amount, to fall by half • **persistence** → does not break down / stays in the environment for a long time **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues	3
04.4	**1 mark each** 1 mark for each of the 3 points below. **1** Alpha particles have very low penetrating power and are absorbed by clothing / dead skin so external sources cause little exposure **1** Inside the body the alpha particles are absorbed in a very small mass of tissue so damage is concentrated (high RBE - weighting factor 20) **1** Long half-life means it continues to emit for the rest of the person's life / cannot be excreted quickly (contamination not just exposure) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **half-life** → the time for the activity, or the amount, to fall by half **Do not credit** • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues	3
04.5	1 mark for the point below. **1** A previously stable nucleus that becomes radioactive after absorbing a neutron (eg in reactor structures), later decaying to emit ionising radiation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ionising radiation** → radiation with enough energy to remove electrons and form ions	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 299-301; AQA spec 3.4.3.2.14.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	1 mark for the point below. **1** The time taken for half of a pollutant (released into the environment) to break down / degrade **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **half-life** → the time for the activity, or the amount, to fall by half **Do not credit** • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues	1
05.2	**1 mark each** 1 mark for each of the 3 points below. **1** DDT is liposoluble so it is stored in fat / can pass through phospholipid cell membranes and is not excreted (nitrates are water soluble and excreted easily) **1** DDT is persistent (long half-life) so it is not broken down before it is passed to the next trophic level **1** So DDT bioaccumulates in each organism and becomes more concentrated at each successive trophic level **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time • **half-life** → the time for the activity, or the amount, to fall by half • **trophic level** → the feeding level in a food chain **Do not credit** • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3
05.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** The gas is denser than air so it settled close to the ground / did not rise and disperse **1** Low wind velocity (at night) meant it was not dispersed / diluted **1** Short half-life so it broke down before travelling far **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **half-life** → the time for the activity, or the amount, to fall by half **Do not credit** • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues	2

Do not write outside the box

Question	Marking guidance	Marks
05.4	**1 mark each** 1 mark for each of the 2 points below. **1** High solubility means nitrates are easily dispersed / leached / carried by water so they affect a larger area (eg rivers, aquifers, the sea) **1** But dispersal dilutes the pollutant so the concentration at any one point may be lower **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leached** → nutrients or ions washed down out of the soil / washed away by water	2
05.5	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Noise / sound **1** Heat / thermal pollution **1** Ionising radiation **1** Light **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Ionising radiation** → radiation with enough energy to remove electrons and form ions **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240; AQA spec 3.4.1.	
06.1	**1 mark each** 1 mark for each of the 3 points below. **1** The water was recharged during or before the last ice age (10 000-20 000 years ago) when the climate was wetter **1** Current recharge is little or none so any abstraction reduces the stored volume **1** The large volume means shortages take a long time to appear, but eventually the supply runs out – ancient water is often saline, causing soil salinisation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abstraction** → water taken out of a river, lake or aquifer • **aquifer** → a permeable rock that stores groundwater / underground water store • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3

Do not write outside the box

Question	Marking guidance	Marks
06.2	**1 mark each** 1 mark for each of the 3 points below. **1** Two satellites whose orbits are affected by the force of gravity **1** The mass of water in aquifers below changes gravity, so changes in the orbit / distance between the satellites indicate changes in groundwater mass **1 Advantage:** the whole Earth is surveyed every 30 days, whereas borehole water-table measurements are slow and hard to coordinate over large areas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store	3
06.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Low-pressure irrigation pipes to reduce leaks **1** Growing crops that need less water than wheat / maize **1** Charging consumers by volume used (metering) **1** Inter-basin transfer of water 450 km from the Yangtze **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
06.4	1 mark for the point below. **1** Trees intercept rainfall and transpire large volumes, reducing soil moisture and the water percolating down to the aquifer **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 119-121; AQA spec 3.2.2.1-3.2.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**1 mark each** 1 mark for each of the 3 points below. **1** The climate naturally fluctuates (solar output, Earth's orbit) so extreme events can occur without human influence **1** An individual event, or even several, does not show a trend – the change may be in the frequency of events which can only be assessed years later **1** Changes occur at different time and spatial scales (local storm vs global trend) – interconnected systems make attribution complex **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **anthropogenic** → human-caused / caused by human activity	3
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Towns have a heat-island effect (heat emissions, dark surfaces absorbing light) **1** As the town grows around the weather station the warming recorded is partly local, not global – older equipment / sparse global coverage also reduce reliability **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
07.3	**Marks shared** Up to 5 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2 Tree rings:** width of each annual ring shows the growth rate, indicating temperature / rainfall in that year – rings easily dated by counting – limitation: growth also affected by other factors (water, pests, shading), only covers the life of the tree / limited regions **up to 2 Ice cores:** annual snow layers compact into ice – air bubbles trapped in the ice give the CO2 (and methane) concentration when trapped – oxygen isotope ratios indicate temperature – up to 800 000 years (Dome C) – limitation: deep layers compressed / less distinct, dating needs radio-isotope analysis and volcanic ash layers – only polar regions **up to 2 Others accepted:** coral growth rings – pollen in lake sediments dated by radiocarbon **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	5
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 100-102; AQA spec 3.2.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Σn(n - 1) = (40 x 39) + (30 x 29) + (20 x 19) + (10 x 9) = 1560 + 870 + 380 + 90 = 2900 **1** N(N - 1) = 100 x 99 = 9900 **1** D = 9900 / 2900 = 3.41 (accept 3.4) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • **Simpson** → a diversity index that uses both the number of species and their abundance **Do not credit** • A bare wrong answer with no working. • 'more species always means a higher index' - the evenness of the abundances matters as well	3
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** Woodland A has a higher biodiversity (greater variety and more even abundance) than woodland B **1** The index does not identify which species are present / a rare or important species may be missed – depends on identification skill and sampling method **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **biodiversity** → the variety of life: habitat, species and genetic diversity	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402; AQA spec 3.7.2.3.	
09.1	**1 mark each** 1 mark for each of the 2 points below. **1** Predicting the route / movement a pollutant will take through the environment (based on its properties and features of the environment) **1** To assess where it will become concentrated / which locations and organisms are most at risk, so monitoring can focus on those sites **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
09.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Sea current velocity and direction **1 Properties of the radionuclides:** half-life, solubility, tendency to adsorb onto sediments / bioaccumulate **1** Food chain pathways, eg uptake by seaweed, shellfish, fish eaten by people **1** Geology / hydrology of the coast – sediment deposition areas (mud flats) **1 Human activities in the area:** fishing, seaweed collection, beach use **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **half-life** → the time for the activity, or the amount, to fall by half • **adsorb** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3
09.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Identify the group of people whose lifestyle gives them the highest exposure (eg people who eat a lot of locally caught shellfish / fish, live near the mud flats, collect seaweed) **1** Monitor their exposure / the concentration of radioactive material in the foods they eat or their body **1** If their exposure / risk is acceptably low, everyone else is assumed to be at even lower risk – if not, emissions are controlled **1** CGM assesses exposure before health effects occur (not used to detect damage) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Critical Group** → the people most at risk / most heavily exposed	3

Do not write outside the box

Question	Marking guidance	Marks
09.4	**Any 2 of 7** 1 mark each for any 2 of the 7 points below, in any order, maximum 2. **1** Seawater **1** Sediment / mud **1** Seaweed **1** Shellfish / molluscs / fish **1** Milk, vegetables or meat from nearby farms **1** Atmospheric dust **1** Soil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243, 303-304; AQA spec 3.4.3.1.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • Organic nutrients (sewage, manure, silage effluent, paper mill / tannery / food processing effluent) are food for aerobic microbes whose respiration deoxygenates the water, killing fish and invertebrates – bacterial growths cover the bed, blocking light • Pathogens in sewage (cholera, typhoid, dysentery) spread via contaminated water – decay of organic matter releases inorganic nutrients • **Inorganic nutrients:** nitrates (fertiliser / manure runoff, leaching) and phosphates (sewage effluent, detergents, eroded soil) cause cultural eutrophication: algal blooms, shading of macrophytes, cyanobacterial toxins, night-time and end-of-summer deoxygenation • **Human health:** nitrites and methaemoglobinaemia – possible carcinogen via nitrosamines • **Control of organic wastes:** sewage treatment (pre-treatment, primary sedimentation removing >95% organic matter, aeration / activated sludge or trickling filters, anaerobic digestion of sludge with biogas) • **Control of nutrients:** tertiary phosphate stripping with iron(III) sulfate, denitrification in filter beds, micro-strainers / UV – Nitrate Vulnerable Zones, fertiliser timing / amount, low-solubility fertilisers, legumes, manure management – dredging sediments • **Evaluation:** sewage treatment has virtually eliminated deoxygenation and waterborne disease in developed countries but storm overflows and combined sewers still discharge untreated sewage – tertiary treatment is expensive so only used at sensitive sites – diffuse agricultural nitrate is hard to control and aquifers respond slowly – phosphate legacy in sediments • Judgement on point-source (sewage) control being more successful than diffuse-source (agriculture) control	9

Do not write outside the box

Question	Marking guidance	Marks
	Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **leaching** → nutrients or ions washed down out of the soil / washed away by water • **eutrophication** → nutrient enrichment of water leading to algal growth • **deoxygenation** → loss of dissolved oxygen from the water **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	
	Model answer - Level 3 (9 marks)	
	Organic wastes - sewage, manure, silage effluent, paper-mill and food-processing effluent - are food for aerobic bacteria, whose respiration strips dissolved oxygen from the water so fish and invertebrates suffocate and bacterial growths smother the bed; sewage also carries pathogens such as cholera and typhoid. Inorganic nutrients act differently: nitrate leached from fertiliser and manure and phosphate from sewage, detergents and eroded soil cause cultural eutrophication, in which algal blooms shade macrophytes, cyanobacteria release toxins and night-time respiration and decay deoxygenate the water. Nitrate in drinking water can cause methaemoglobinaemia and may form carcinogenic nitrosamines.	
	Point sources are controlled by sewage treatment: screening and grit removal, primary sedimentation removing most organic solids, aeration with activated sludge or trickling filters, and anaerobic digestion of sludge yielding biogas. Nutrients need tertiary treatment - phosphate stripping with iron(III) sulfate, denitrifying filter beds, micro-strainers and UV. Diffuse agricultural sources are managed by Nitrate Vulnerable Zones, limits on the timing and amount of fertiliser, low-solubility fertilisers, legumes, manure management and buffer strips.	
	Sewage treatment has virtually eliminated river deoxygenation and waterborne disease in developed countries, but storm overflows on combined sewers still discharge untreated sewage, and tertiary treatment is expensive so it is used only where eutrophication is a risk. Diffuse nitrate from farmland is much harder to control because it comes from thousands of sources and depends on farmers' behaviour and the weather. Control of point sources has therefore been far more successful than control of diffuse ones.	
	Weaker answer - Level 1 (3 marks)	
	Sewage in rivers uses up the oxygen because bacteria break it down, so fish die. Fertilisers wash into rivers and cause eutrophication, where algae grow and block the light and then the water runs out of oxygen. Sewage is treated in sewage works to clean it before it goes into rivers. Farmers should use less fertiliser. Sewage treatment works well but fertiliser is still a problem.	
	What makes the difference	
	• Level 1 has the outline of both mechanisms but Level 3 explains them precisely (aerobic respiration, shading, night-time respiration, pathogens, health effects) and names the pollutant sources.	
	• Level 3 describes the stages of sewage treatment and the specific nutrient controls; Level 1 says 'sewage works' and 'less fertiliser'.	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 3's evaluation gives reasons (storm overflows, cost of tertiary treatment, diffuse sources) for its point-vs-diffuse judgement; Level 1 asserts it.	
10.2	1 mark for the point below. **1** Algae / cyanobacteria absorb nutrients directly from the water (not the sediments) so they use dissolved nutrients more easily and grow rapidly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **eutrophication** → nutrient enrichment of water leading to algal growth **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 266-272; AQA spec 3.4.3.2.9.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Properties:** state of matter, density, persistence, toxicity, specificity, reactivity, adsorption, solubility, bioaccumulation / biomagnification, synergism, mutagenic / carcinogenic / teratogenic action, mobility • **How properties predict behaviour:** dispersal (gases vs solids, density - Bhopal), how far and how long (persistence, mobility), where they concentrate (adsorption on sediments, food chains) • Environmental features interacting with properties: temperature inversions, currents, pH, oxygen, light, adsorbent materials • **Control based on properties:** persistent / liposoluble pesticides banned (organochlorines) and replaced by non-persistent / specific ones – systemic pesticides and seed coatings • **Heavy metals:** lead uses discontinued, storage at high pH to reduce solubility – mercury chemical form – radioactive waste categorised by half-life / activity and contained (encapsulation, sealed storage) • Reactive primary pollutants controlled to prevent secondary pollutants (catalytic converters, FGD, urea sprays) – oil properties (floats, insoluble) exploited by booms and skimmers	25

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Question	Marking guidance	Marks
	– density in cyclone separators • **Principles:** CPA and CGM based on properties – emission location / timing – ALARA and BATNEEC – polluter pays – precautionary principle • **New technologies:** satellite oil-spill monitoring, GPS, polymer adsorption, phytoremediation, bioremediation • **Evaluation:** prevention better than cure – limits of remediation – cost-effectiveness – contrast success of ozone (Montreal) with climate change – judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-245, 262-266, 275-285, 295-304; AQA spec 3.4.1-3.4.3.	
	Model answer - Level 5 (22 marks)	
	Pollutants cause damage in proportion to how much reaches sensitive organisms and how harmful it is when it gets there. Both depend on properties, so knowing a pollutant's properties lets us predict where it will go and which control will work.	
	State of matter and density decide dispersal. Solid particulates settle near the source unless they are very small: PM10 and smaller stay suspended, travel far and penetrate lungs, so smoke is controlled at the point of release with electrostatic precipitators, cyclones and bag filters, and smokeless zones prevent its production. Gases disperse widely, and dense gases such as the hydrogen cyanide released at Bhopal hug the ground, so location and emission height matter. Persistence decides whether dilution is acceptable. Sewage and other biodegradable wastes break down, so treatment plus dilution in a river or the sea is sufficient; organochlorine pesticides, CFCs, heavy metals and long-lived radioisotopes are not broken down, so they travel, accumulate and must be contained, banned or replaced - DDT and CFCs were both controlled by substitution once their persistence was understood.	
	Solubility governs the pathway. Water-soluble nitrate leaches into rivers and aquifers, so it is controlled by nitrate vulnerable zones, fertiliser timing and buffer strips; liposoluble pollutants such as DDT and methyl mercury pass into fat and are biomagnified along food chains, as at Minamata, so even low concentrations in water are dangerous and the only effective control is to stop release. Adsorption on clay immobilises caesium and metals, which is why radioactive discharges are monitored in sediments and why activated carbon and polymer adsorbents are used deliberately in effluent treatment.	

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Question	Marking guidance	Marks
	Toxicity, specificity and mode of action set acceptable doses and identify species at risk. Neonicotinoids are toxic to insects but not vertebrates because they cannot cross the vertebrate blood-brain barrier, which explains both their popularity and their effect on bees; pyrethroids replaced organochlorines because they are less persistent. For mutagens and carcinogens such as ionising radiation there is no safe dose, so exposure is kept As Low As Reasonably Achievable with shielding, distance, time limits and monitoring. Reactivity explains secondary pollutants - sulfur dioxide oxidised to sulfuric acid, nitrogen oxides and hydrocarbons forming ozone and PANs in photochemical smog - so control targets the primary emissions with flue-gas desulfurisation and catalytic converters. Synergism between smoke and sulfur dioxide, or ozone and sulfur dioxide, means limits must consider pollutants together.	
	Properties interact with the environment: temperature inversions trap pollutants, currents disperse them, pH and oxygen alter solubility. Critical Pathway Analysis combines the two to predict where a discharge will concentrate and to focus monitoring, and Critical Group Monitoring checks the most exposed people. These principles have reduced smoke, sulfur dioxide, lead and organochlorines dramatically in developed countries. Their limits are pollutants with diffuse sources or that underpin economies - nitrate, carbon dioxide - where knowing the properties is not the problem; changing behaviour is. Even there, the precautionary principle applies the same logic: predict the properties of a new substance from similar ones and control it before release, because prevention is always cheaper than remediation.	
	What makes the difference	
	• Every paragraph is organised around a **property**, each linked to a prediction and a named control - the essay answers the question throughout rather than describing pollutants.	
	• Examples span the specification (smoke, acid rain, pesticides, heavy metals, radiation, nutrients, oil), with quantitative and named detail (Bhopal, Minamata, PM10, ALARA), which is what Level 5 breadth means.	
	• The final paragraph evaluates: where the approach has worked, where it has not and why, and reaches a justified conclusion.	
	• A Level 3 version of this essay would describe several pollutants and their effects accurately but only occasionally connect a property to a control, and would end without a judgement.	

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Question	Marking guidance	Marks
11.2	**To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society?** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Fossil fuel features that shaped society: energy density, storage, transport, high temperatures, liquid vehicle fuels, established infrastructure • **Renewable features:** intermittent / unpredictable (wind, solar) vs predictable (tidal) vs controllable (HEP, biofuel, geothermal) – low energy density – location – low pollution but manufacturing impacts • Technologies and developments for each resource with examples – new technologies improving usability • **Storage and grid management:** pumped storage, batteries, hydrogen economy, P2G, V2G, thermal storage, peak shaving – the role of electricity as a secondary fuel • **Economics and politics:** development costs, interest, subsidies, external costs of fossil fuels, falling renewable costs • Energy conservation as part of the solution – sustainability principles - natural systems run on low-density renewable energy • **Judgement:** electricity largely yes	25
	– heat, aviation, shipping and high-temperature industry harder – transition period with nuclear / CCS *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	

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